

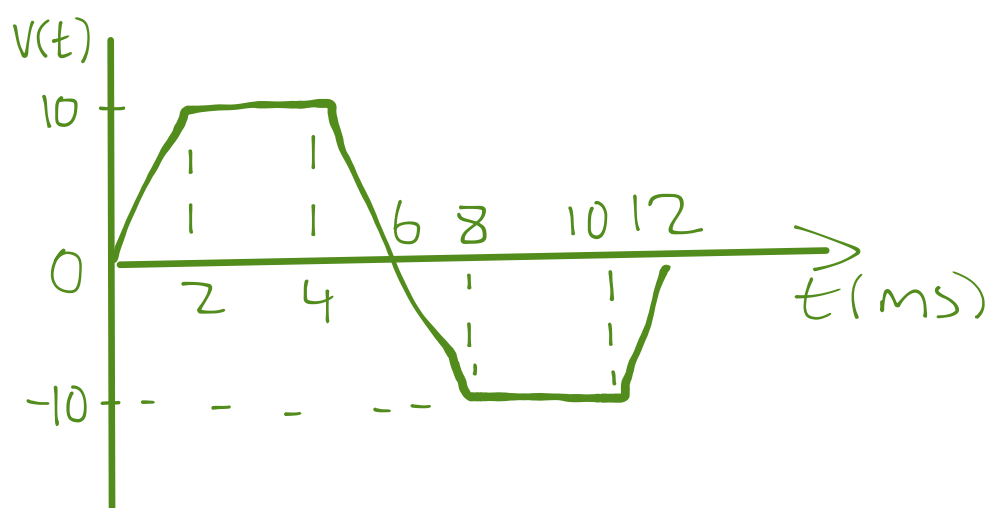
Tutorial 8. Memo

Questions: 6.6 ; 6.10 ; 6.12 ; 6.13 ; 6.17 ;
6.40 ; 6.42 ; 6.46 ; 6.48 ; 6.54

Solutions:

6.6 $V(t)$ waveform applied across a
55-mF capacitor.

→ Draw the current waveform



$$i = C \underbrace{\frac{dV}{dt}}_{\text{slope of waveform}}$$

$$\text{for } 0 < t < 2: \frac{dV}{dt} = \frac{10 - 0}{(2 \times 10^{-3}) - 0} = 5000$$

$$\Rightarrow \dot{i} = C \frac{dV}{dt} = 55 \mu(5000) \\ = \underline{275 \text{ mA}}$$

$$\text{for } 2 < t < 4: \frac{dV}{dt} = \frac{10-10}{(4-2) \times 10^{-3}} = 0$$

$$\therefore \dot{i} = 0$$

Note for $t = 8 - 10 \text{ ms}$, also has a horizontal slope

$$\therefore \dot{i} = 0$$

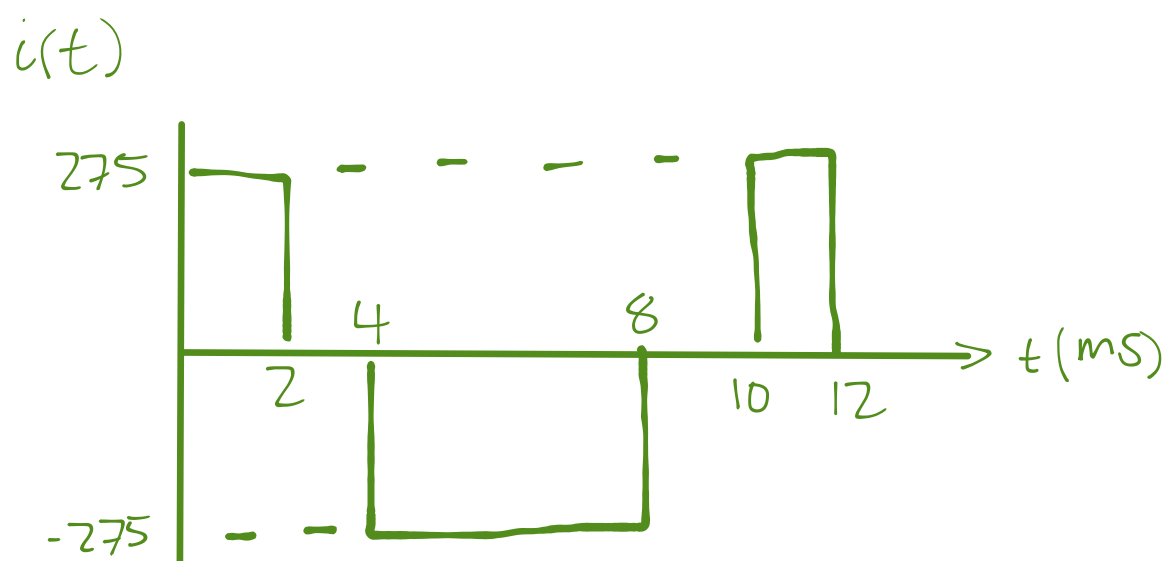
$$\text{for } 4 < t < 8: \frac{dV}{dt} = \frac{-10-10}{(8-4) \times 10^{-3}} \\ = -5000$$

$$\Rightarrow \dot{i} = -275 \text{ mA}$$

$$\text{for } 10 < t < 12: \frac{dV}{dt} = \frac{0-(-10)}{(12-10) \times 10^{-3}} \\ = 5000$$

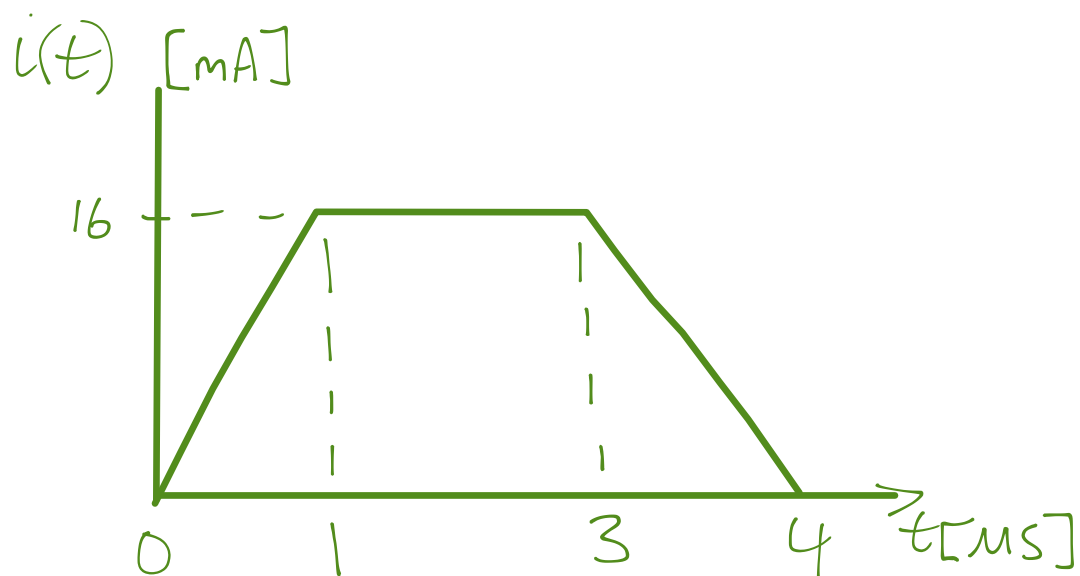
$$\Rightarrow \dot{i} = 275 \text{ mA}$$

Current waveform:



6.10 Voltage wave applied across 5 mF capacitor.

Determine current through capacitor.



$$\Rightarrow i(t) = C \frac{dv}{dt}$$

→ Using the approach from 6.6, the solution could be derived.

$$V = \begin{cases} 16t, & 0 < t < 1\text{ms} \\ 16, & 1 < t < 3\text{ms} \\ 64 - 16t, & 3 < t < 4\text{ms} \end{cases}$$

$$\frac{dV}{dt} = \begin{cases} 16\text{M} & , \quad 0 < t < 1\text{ms} \\ 0 & , \quad 1 < t < 3\text{ms} \\ -16\text{M} & , \quad 3 < t < 4\text{ms} \end{cases}$$

$$\Rightarrow i(t) = \begin{cases} 80\text{nA}, & 0 < t < 1\text{ms} \\ 0 & , \quad 1 < t < 3\text{ms} \\ -80\text{nA}, & 3 < t < 4\text{ms} \end{cases}$$

6.12 $45e^{-2000t}$ V applied across

100 mF capacitor and 12 Ω resistor in parallel.

$\Rightarrow$ calculate power absorbed.

Solution: $i_R = \frac{V}{R} = \frac{45e^{-2000t}}{12} = 3.75e^{-2000t}$

$$\begin{aligned} i_C &= C \frac{dV}{dt} = (100 \times 10^{-3}) \times 45(-2000)e^{-2000t} \\ &= -9000e^{-2000t} \end{aligned}$$

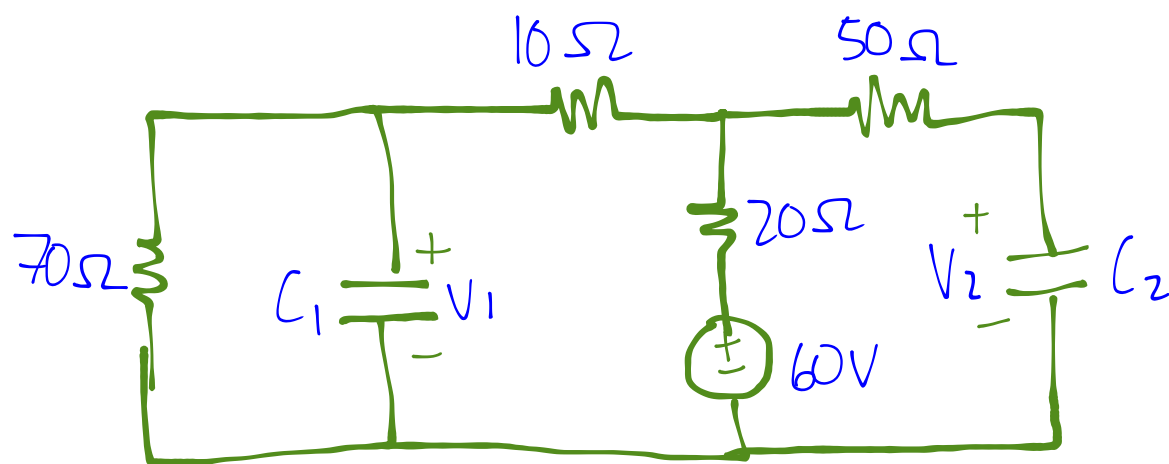
$\hookrightarrow$ Kirchhoff current law

$$i = i_R + i_C = -8996.25e^{-2000t}$$

$$\Rightarrow p = Vi = -404831.25e^{-4000t} \text{ W}$$

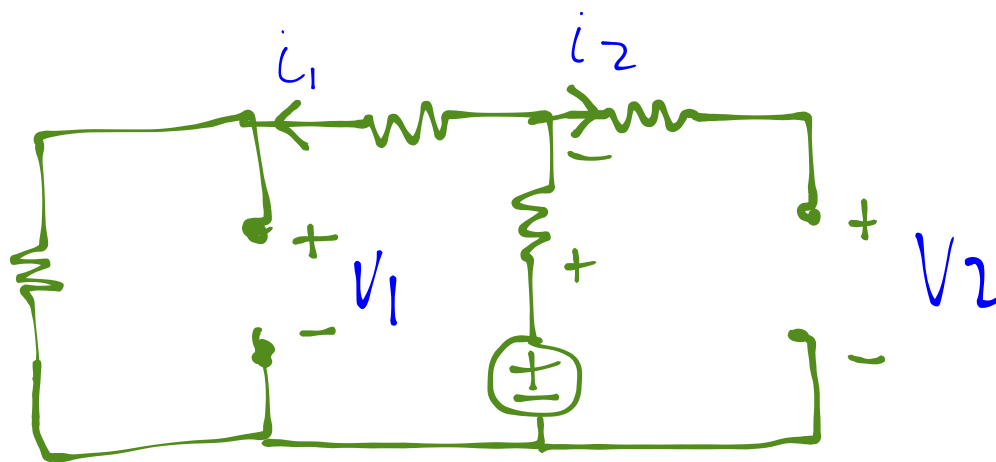
6.13 Find Voltage across the capacitors

→ Assume DC conditions



→ Under DC conditions, capacitors can be seen as opencircuits.

→ redraw circuit:



$$i_2 = 0$$

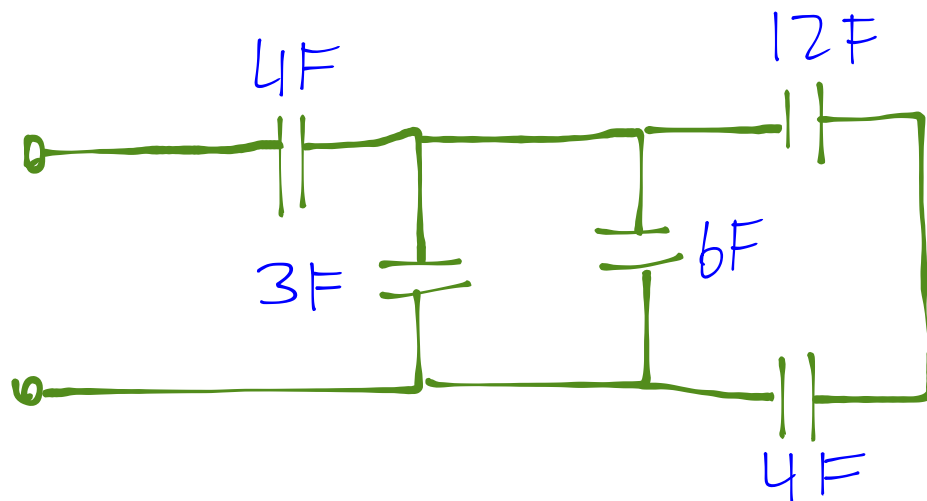
$$i_1 = \frac{V}{R} = \frac{60}{70+10+20} = 0,6 \text{ A}$$

$$\rightarrow V_1 = i_1 \times 70 = \underline{42 \text{ V}}$$

$$\rightarrow V_2 = 60 - 20i_1 = \underline{72 \text{ V}}$$

6.17 Determine equivalent capacitance

a)



4F in series with 12F

$$\therefore C_s = \frac{4 \times 12}{4 + 12} = 3F$$

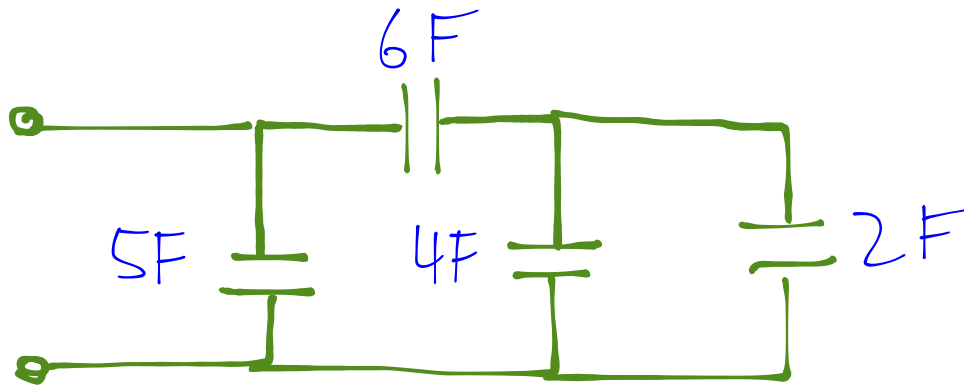
3F in parallel with 6F, in addition to C_s

$$\therefore C_{||} = 3 + 6 + C_s \\ = 12F$$

4F in series with $C_{||}$

$$\therefore C_{eq} = \frac{4 \times C_{||}}{4 + C_{||}} = 3F \rightarrow$$

b)



4F in parallel with 2F

$$C_{//} = 4 + 2 = 6F$$

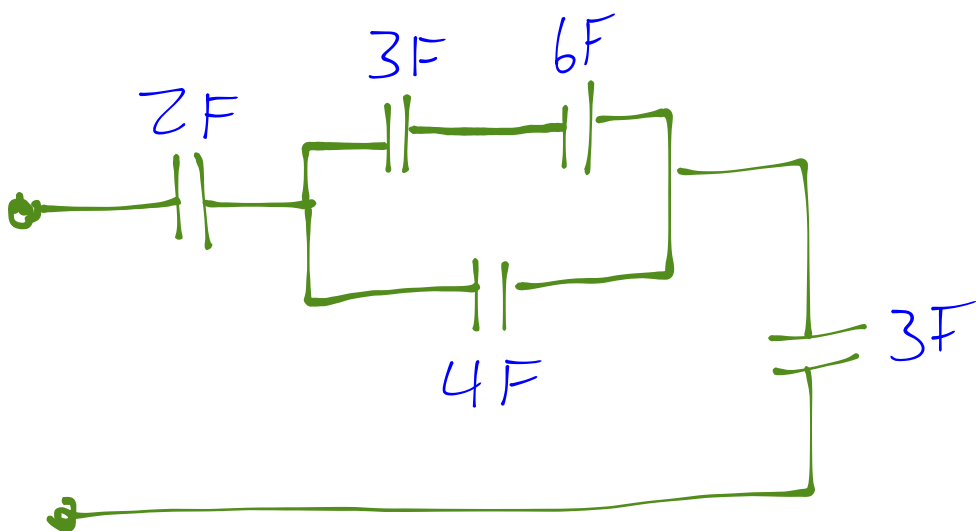
6F in series with $C_{//}$

$$C_s = \frac{6 \times C_{//}}{6 + C_{//}} = 3F$$

C_s in parallel to 5F

$$C_{eq} = 5 + C_s = 8F \rightarrow$$

c)



3F and 6F in parallel to 4F

$$C_{//} = \frac{3 \times 6}{3 + 6} + 4 = 6 \text{ F}$$

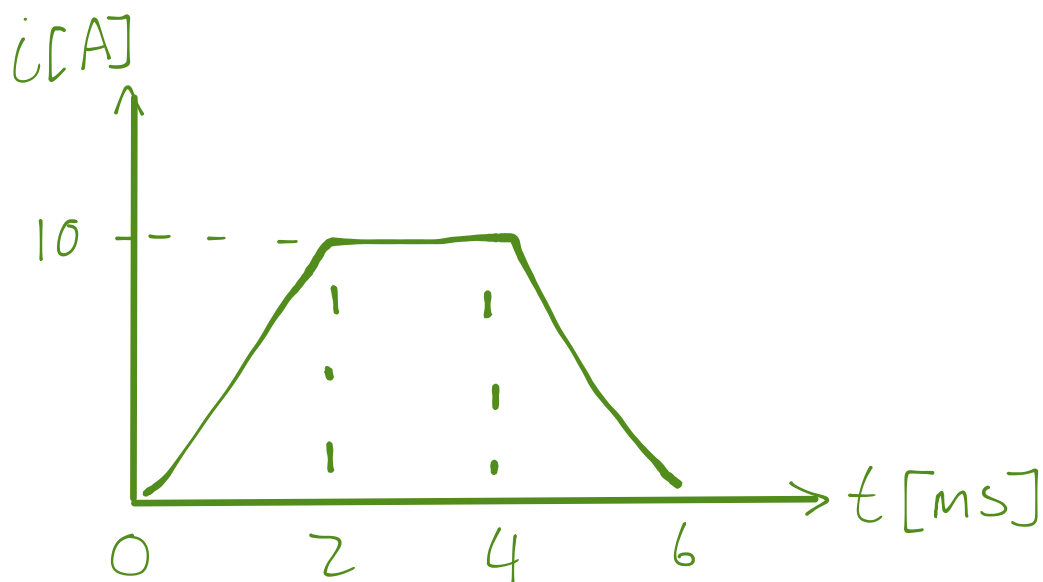
2F in series with $C_{//}$ and 3F

$$\frac{1}{C_{eq}} = \frac{1}{2} + \frac{1}{C_{//}} + \frac{1}{3}$$

$$C_{eq} = 1 \text{ F}$$

6.40 Current through 5mH inductor.

Determine the voltage at $t = 1, 3$ and 5 ms .



$$i = \begin{cases} 5t & , 0 < t < 2 \text{ ms} \\ 10 & , 2 < t < 4 \text{ ms} \\ 30 - 5t & , 4 < t < 6 \text{ ms} \end{cases}$$

$$\Rightarrow V = L \frac{di}{dt} = (5 \times 10^{-3}) \times \begin{cases} 5000, & 0 < t < 2 \text{ ms} \\ 0, & 2 < t < 4 \text{ ms} \\ -5000, & 4 < t < 6 \text{ ms} \end{cases}$$

$$\text{at } t = 1 \text{ ms} \Rightarrow V = (5 \times 10^{-3}) \times 5000 = \underline{25 \text{ V}}$$

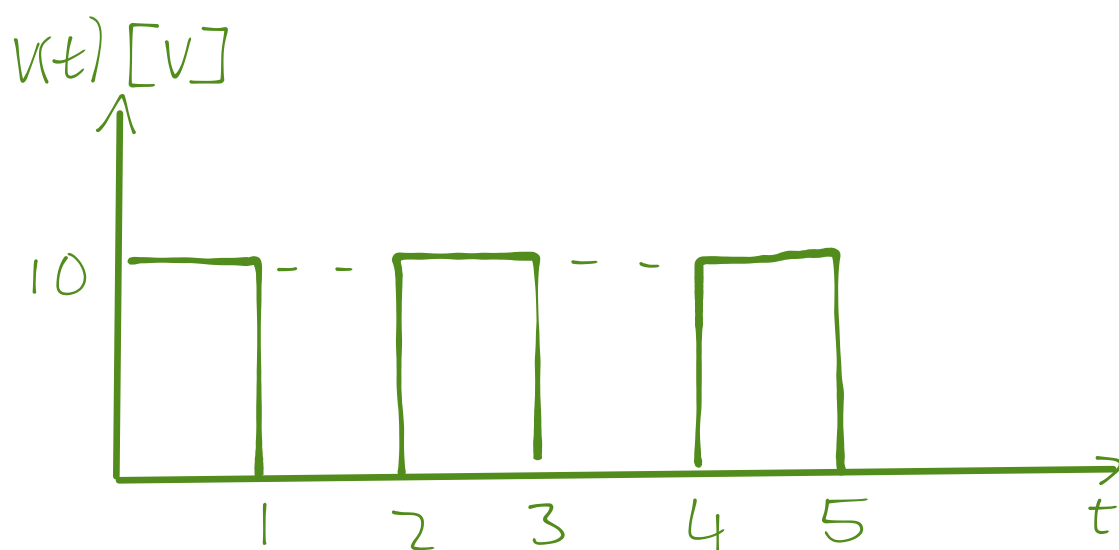
$$\text{at } t = 3 \text{ ms} \Rightarrow \underline{V = 0 \text{ V}}$$

$$\text{at } t = 5 \text{ ms} \Rightarrow (5 \times 10^{-3}) \times -5000 = \underline{-25}$$

6.42 Voltage wave applied across
5H inductor.

Calculate current through inductor.

→ Assume $i(0) = -1 \text{ A}$



$$i = \frac{1}{L} \int_0^t v dt + i(0)$$

$$= \frac{1}{5} \int_0^t v(t) dt - 1$$

For $0 < t < 1$: $i = \frac{1}{5} \int_0^t 10 dt - 1$

$$= 2t - 1 \text{ A}$$

For $1 < t < 2$: $\dot{i} = 0 + i(1) = 1 \text{ A}$

For $2 < t < 3$: $i = \frac{1}{5} \int 10 dt + i(2)$

$$= 2t \Big|_2^t + 1$$

$$= 2t - 4 + 1$$

$$= 2t - 3 \text{ A}$$

For $3 < t < 4$: $i = 0 + i(3) = 3 \text{ A}$

For $4 < t < 5$: $i = \frac{1}{5} \int_4^t 10 dt + i(4)$

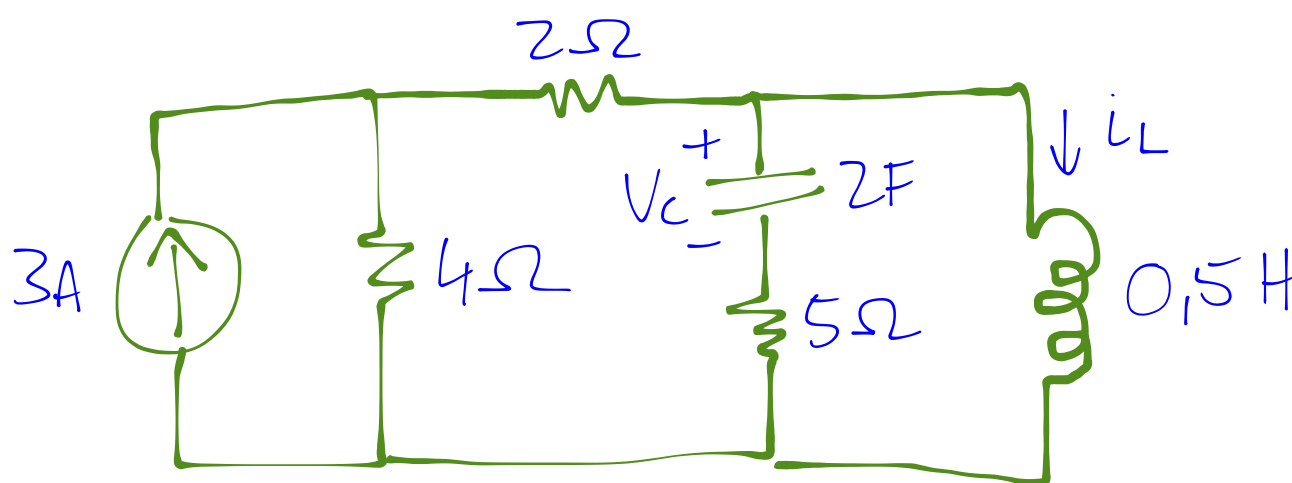
$$= 2t \Big|_4^t + 3$$

$$= 2t - 5 \text{ A}$$

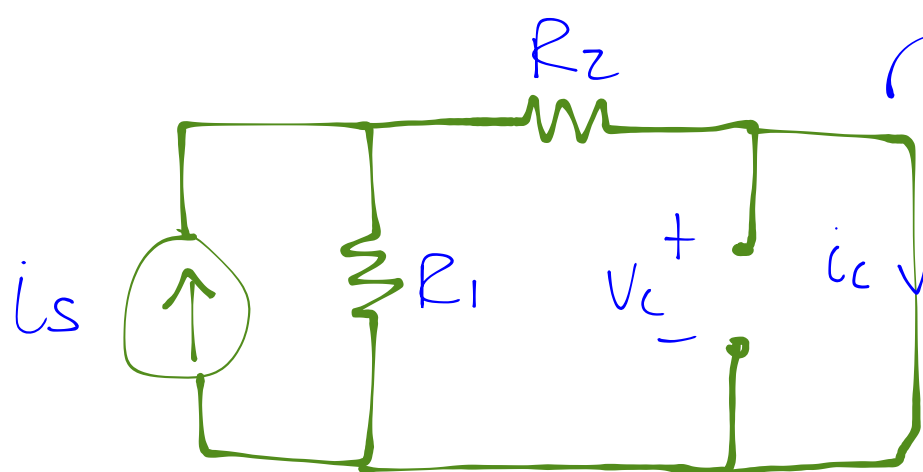
$$\therefore i(t) = \begin{cases} 2t - 1A, & 0 < t < 1 \\ 1A, & 1 < t < 2 \\ 2t - 3A, & 2 < t < 3 \\ 3A, & 3 < t < 4 \\ 2t - 5, & 4 < t < 5 \end{cases}$$

6.46 Find V_C , i_L and the energy stored in the capacitor and inductor.

→ Assume DC conditions



Redraw circuit as seen under DC conditions



$$V_L = 0V$$

Apply current division:

$$\begin{aligned} i_L &= i_s \left(\frac{R_1}{R_1 + R_2} \right) \\ &= 3 \left(\frac{4}{4 + 2} \right) = 2A \end{aligned}$$

Calculate energy for inductor:

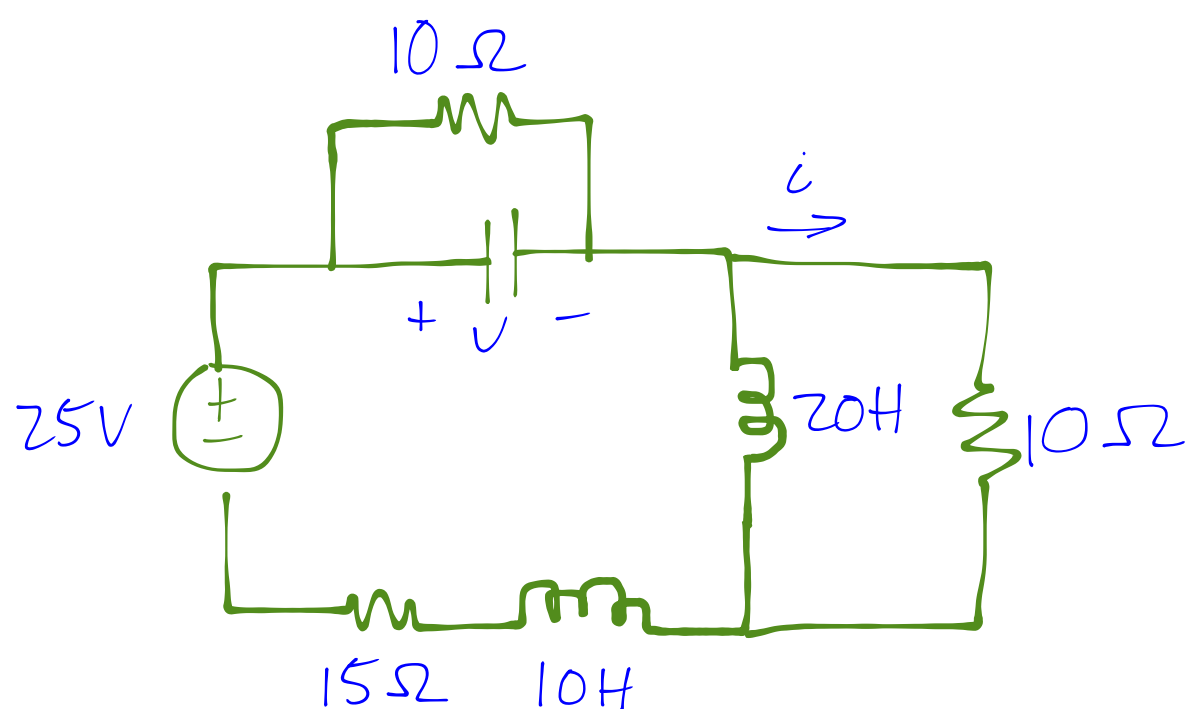
$$\begin{aligned} W_L &= \frac{1}{2} L i_L^2 = \frac{1}{2} (0,5) (2)^2 \\ &= 1J \end{aligned}$$

Calculate energy for capacitor:

$$\begin{aligned} W_C &= \frac{1}{2} C V_c^2 = \frac{1}{2} (2) (0) \\ &= 0J \end{aligned}$$

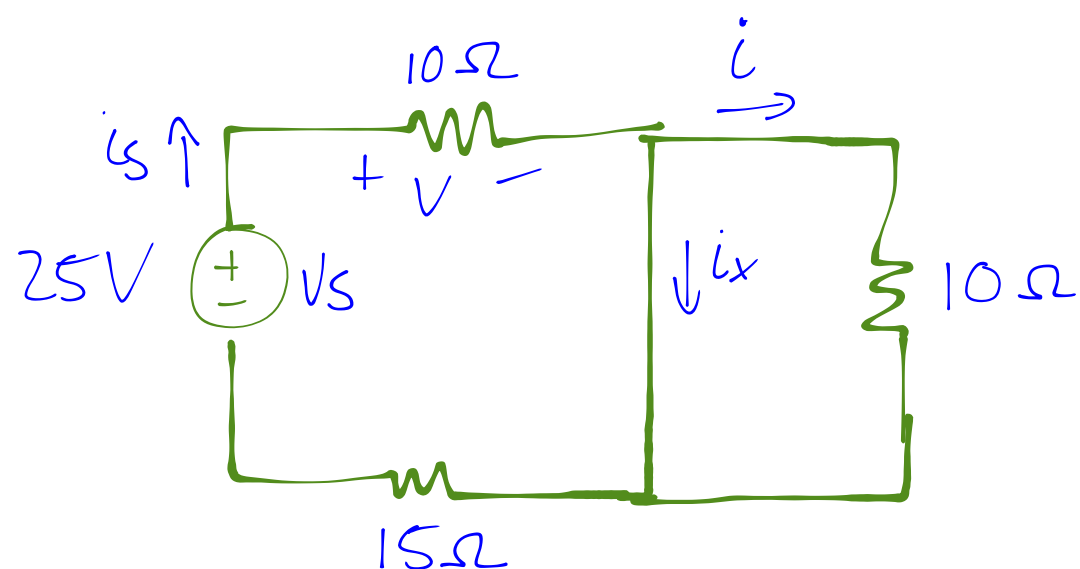
6.48 Find i and V

→ Assume steady state DC conditions



Redraw circuit as seen under

DC conditions:



Current chooses the path of least resistance, therefore all current will flow through i_x rather than i

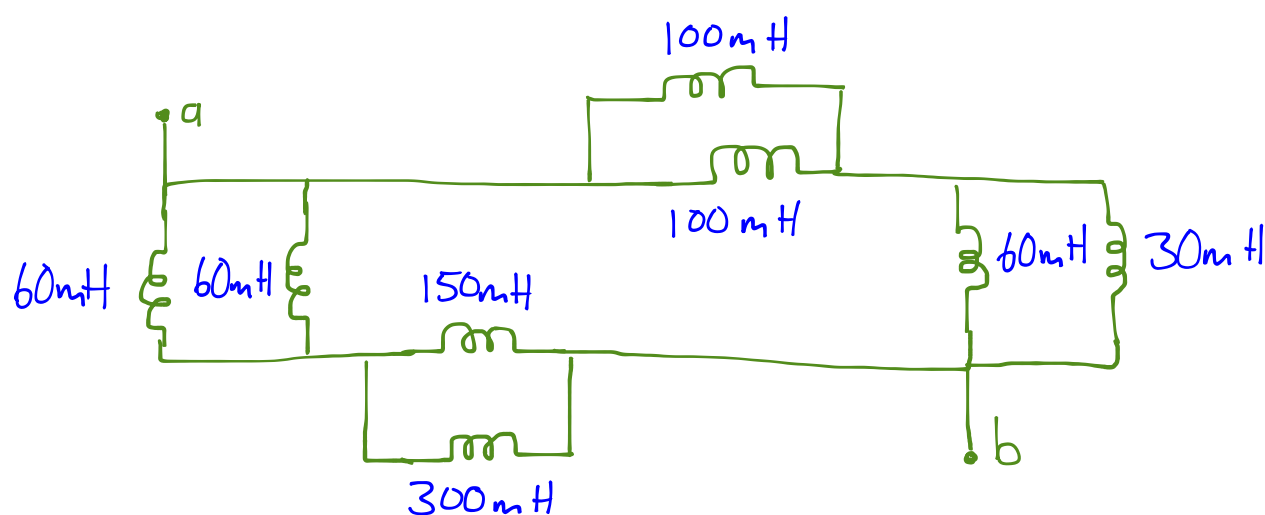
$$\therefore \underline{i = 0 \text{ A}}$$

Calculate i_s using Ohms law

$$i_s = \frac{V_s}{R} = \frac{25}{10+15} = 1 \text{ A}$$

$$\therefore V = 10 \times i_s = 10 \times 1 = 10 \text{ V}$$

6.54 Find equivalent inductance



60mH in parallel to 60mH

$$L_{//1} = \frac{60m \times 60m}{60m + 60m} = 30mH$$

30mH in parallel to 60mH

$$L_{//2} = \frac{30m \times 60m}{30m + 60m} = 20mH$$

100mH in parallel to 100mH

$$L_{//3} = \frac{100m \times 100m}{100m + 100m} = 50mH$$

300mH in parallel to 150mH

$$L_{//4} = \frac{300m \times 150m}{300m + 150m} = 100mH$$

$L_{//3}$ in series with $L_{//2}$

$$\rightarrow L_{S1} = L_{//3} + L_{//2} = 70mH$$

$L_{//1}$ in series with $L_{//4}$

$$\rightarrow L_{S2} = L_{//1} + L_{//4} = 130mH$$

L_{s1} in parallel with L_{s2}

$$L_{eq} = \frac{L_{s1} \times L_{s2}}{L_{s1} + L_{s2}} = \frac{70m \times 130m}{70m + 130m}$$

$$= 45,5mH$$